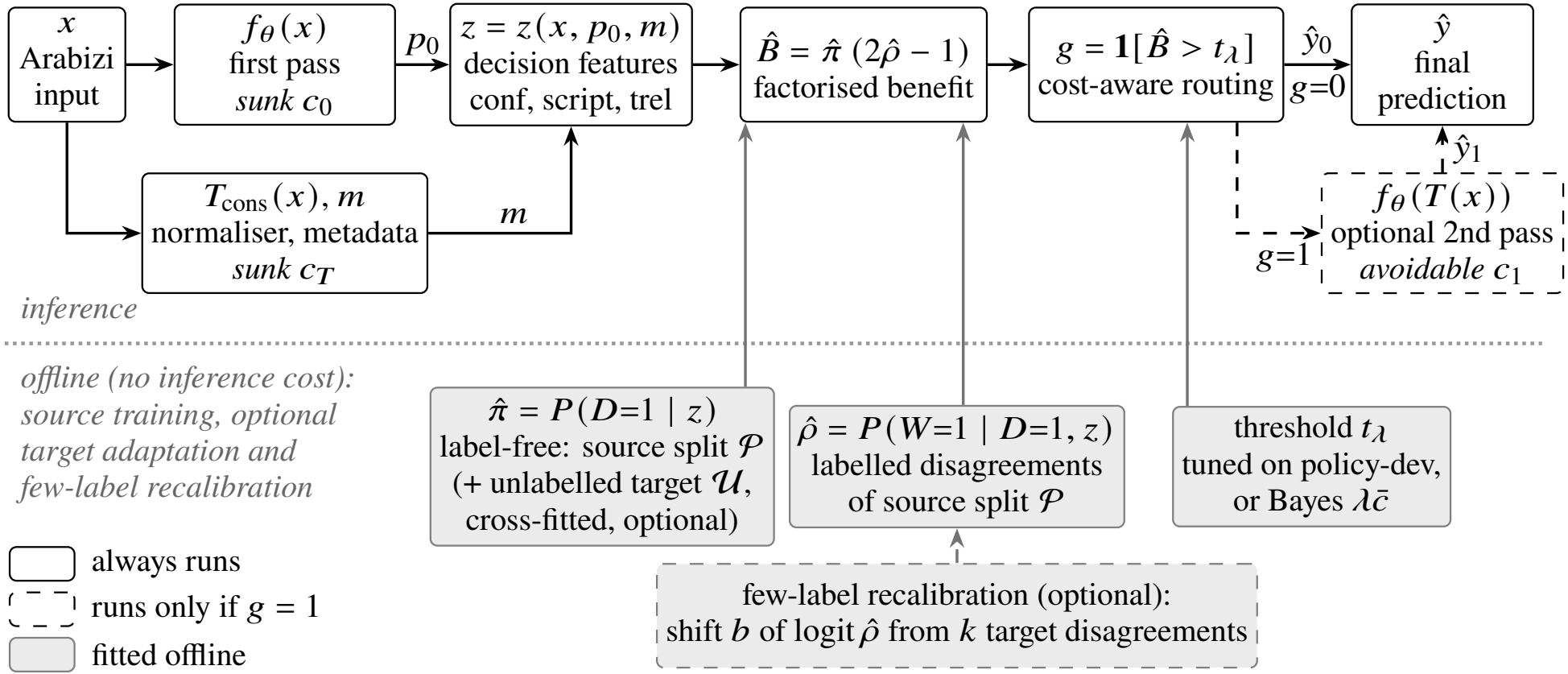

x
Arabizi input
$f_\theta(x)$
first pass
sunk c_0
p_0
$z = z(x, p_0, m)$
decision features
conf, script, trel
$\hat{B} = \hat{\pi}\,(2\hat{\rho} - 1)$
factorised benefit
$g = \mathbf{1}[\hat{B} > t_\lambda]$
cost-aware routing
$\hat{y}_0$
g=0
$\hat{y}$
final prediction
$\hat{y}_1$
$T_{\text{cons}}(x), m$
normaliser, metadata
sunk c_T
m
g=1
$f_\theta(T(x))$
optional 2nd pass
avoidable c_1
inference
offline (no inference cost):
source training, optional
target adaptation and
few-label recalibration
$\hat{\pi} = P(D{=}1 \mid z)$
label-free: source split $\mathcal{P}$
(+ unlabelled target $\mathcal{U}$,
cross-fitted, optional)
$\hat{\rho} = P(W{=}1 \mid D{=}1, z)$
labelled disagreements
of source split $\mathcal{P}$
threshold t_λ
tuned on policy-dev,
or Bayes $\lambda\bar{c}$
few-label recalibration (optional):
shift b of logit $\hat{\rho}$ from k target disagreements
always runs
runs only if $g = 1$
fitted offline